Tower Monitoring Alarm Configuration Guide (Includes Teltonika-IOT, EFOY and Victron Device Configuration)

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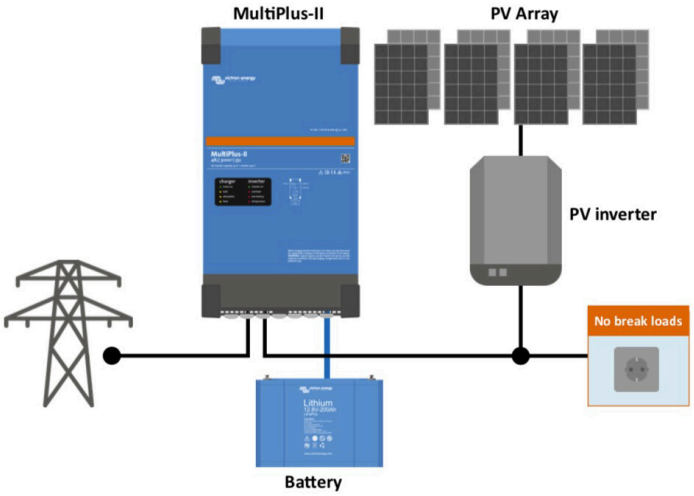
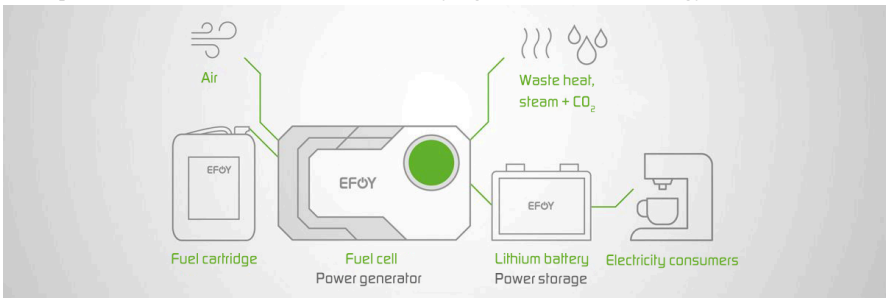
Introduction

When we refer to **Tower Monitoring** in *Genesis*, we are describing an integrated ecosystem of multiple IoT subsystems working together to ensure the tower's **operational continuity, energy efficiency, and communication reliability**.
Each subsystem plays a distinct but interconnected role, contributing to the tower's overall intelligence and autonomy.

- **Victron Energy** oversees **Power and Energy Monitoring**, tracking parameters such as battery voltage, charging current, solar input, and load consumption. This ensures that the energy system operates within safe thresholds and provides early warning of battery or solar irregularities.



- **EFOY Fuel Cells** enable **Fuel Cell Backup Monitoring**, safeguarding critical uptime by reporting fuel levels, voltage output, state of charge, and internal temperature — essential for remote sites relying on autonomous energy sources.
  - 

- **Teltonika IoT Devices** manage **Network Connectivity and Device Health Monitoring**, including mobile network quality, SIM data usage, geofencing, input/output state changes, and advanced jamming detection for communication assurance.
  -

Together, these intelligent systems form a **unified monitoring architecture** within *Genesis*. They continuously stream telemetry data from distributed towers into the Genesis IoT cloud, where it is analyzed, stored, and correlated with configurable alarm rules.

System Overview

Each device — whether it's a **Victron solar/battery controller**, **EFOY fuel cell**, or **Teltonika router** — periodically reports status and sensor values to Genesis using secure communication channels.
These data points are evaluated in real time against pre-defined **alarm thresholds and logic rules**. When a condition breaches the configured limits, Genesis automatically:

1. **Generates event alerts** (using the assigned event codes and group codes).
2. **Notifies the appropriate operator or system**, through respective CMS channels (Evalink, AmWin, Lisa, IMMIX, etc...)
3. **Tracks the event lifecycle**, ensuring alerts are automatically cleared/restored when the parameters return to their normal range.

This end-to-end monitoring pipeline transforms raw device data into actionable intelligence, allowing operators to **detect risks early, respond faster, and maintain consistent tower uptime** across all remote locations.

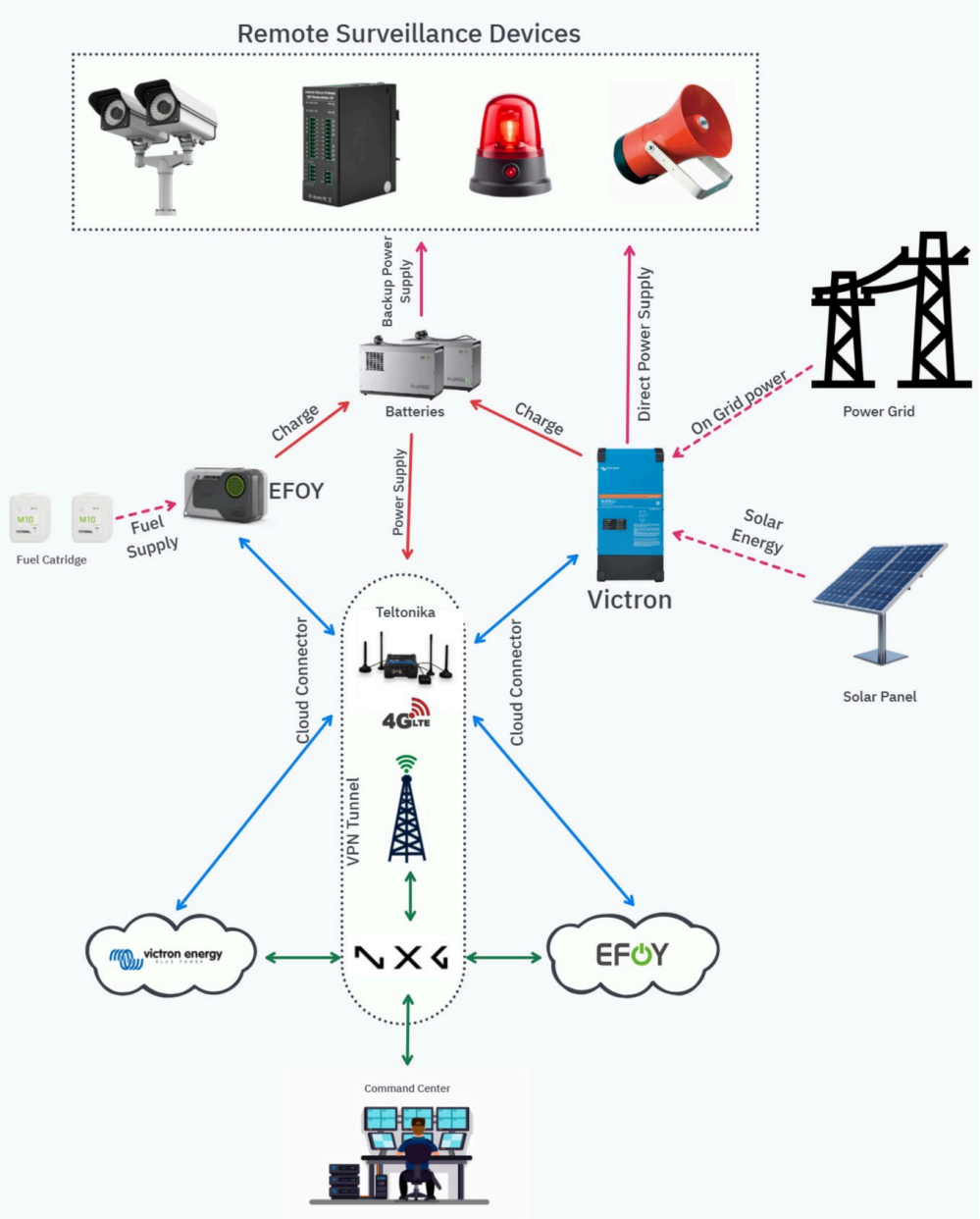
System Architecture

The Tower Monitoring architecture follows a distributed design that enables autonomous operation at remote tower sites while maintaining centralized visibility and control through the Genesis platform.
The diagram below illustrates how the Victron Energy, EFOY Fuel Cell, and Teltonika IoT subsystems interact within the tower ecosystem and how telemetry and alarms are transmitted to the command center.

At the site level, Victron manages solar and battery performance, EFOY ensures backup power availability, and Teltonika provides communication, GPS tracking, and edge telemetry routing. These subsystems communicate securely with their respective cloud services, where Genesis collects real-time operational data through authenticated API channels.

Within Genesis, this incoming data is normalized, correlated, and continuously analyzed against configured alarm rules. Any deviation—such as low battery voltage, reduced solar input, fuel depletion, or network loss—triggers an alarm routed automatically to the operator console or central monitoring system.

This distributed yet unified architecture ensures every tower operates independently in the field while maintaining synchronized health and alarm visibility across all sites in the Genesis command center.



The Tower Monitoring architecture is organized in five layers that describe how field telemetry becomes alarms and operator actions in Genesis.

Field Layer — Devices and Sensors

This layer contains the hardware installed at each tower site and the telemetry each device provides.

- **Victron Energy devices** (MPPTs, Battery Monitors, Inverters) supply power metrics such as battery voltage, charging current, state of charge, solar input, and system alarms. These devices are typically published to the Victron VRM cloud for remote access.
- **EFOY Fuel Cells** report fuel cartridge level, runtime estimates, output voltage, and internal temperature. They usually push data to the EFOY Cloud.
- **Teltonika-IOT routers** handle network connectivity, GPS position, SIM data usage, input/output status, and jamming detection. They provide both status telemetry and remote I/O for site control.

Each device continuously collects sensor data and health indicators that are required for both routine monitoring and alarm detection.

Edge Communication Layer — Local and Mobile Links

This layer covers how devices reach the cloud when fixed infrastructure is not available.

- Victron devices forward telemetry to the Victron VRM service via internet connectivity provided by the site router or cellular uplink.
- EFOY devices forward to the EFOY Cloud using the configured cellular or gateway connection.

- Teltonika routers provide the primary WAN link for the tower, and may also expose local device endpoints (SNMP, HTTP, or vendor APIs) used by on-site systems.

The router also supplies remote access and optional VPN channels. Local I/O states and short term buffering occur here when connectivity is intermittent.

Cloud Aggregation Layer — Vendor Clouds and APIs

Vendor cloud platforms centralize device telemetry and present secure APIs that Genesis consumes.

- **Victron VRM API** provides real time device metrics required for energy monitoring rules.
- **EFOY Cloud API** exposes fuel cell telemetry and status reports.
- **Teltonika Cloud / Local API** supplies router diagnostics, SIM usage statistics, GPS, and I/O states.

Genesis connects to these cloud APIs using tenant level credentials, API keys, or account credentials to retrieve device lists, telemetry streams, and event logs.

Integration Layer — Genesis IoT Engine

Genesis normalizes vendor telemetry and applies alarm logic.

- Genesis discovers devices via cloud API calls and registers them under the tenant and site mapping.
- Incoming telemetry is parsed and mapped to Genesis device models (Energy Monitoring, Fuel Cell Monitoring, Connectivity Monitoring).
- Configurable alarm rules (JSON) evaluate the telemetry against thresholds, delays, and compound logic. Example rule groups include: Battery Voltage, Charger Status, Fuel Level, Network Loss, SIM Overuse, Input State Change, and Jammer Detection.
- When a rule is triggered, Genesis generates an event with assigned event codes, stores related telemetry, and attaches any available verification media.

Monitoring Layer — Command Center and CMS Integration

This layer covers how operators receive and act on alerts.

- Events are routed to operator consoles via integrated CMS channels such as IMMIX, AmWin, Evalink, or LISA.
- Each event lifecycle is tracked: creation, escalation, acknowledgement, and automatic clear/restoration when parameters return to normal.
- Genesis supports operator actions such as remote device commands (e.g., Teltonika remote restart, Victron remote setpoints if supported), notification workflows, and maintenance tickets.

Configuration Overview

In *Genesis*, each tower integrates multiple device types — each with its own configuration logic and threshold rules. The configuration defines how the system interprets sensor values, detects abnormal conditions, and triggers the appropriate alerts.
This section explains how configuration is structured for the three core subsystems: **Victron Energy**, **EFOY Fuel Cell**, and **Teltonika IoT Connectivity**.

Victron Energy Monitoring Configuration

Victron devices act as the **primary power monitoring system** of the tower, focusing on solar charging performance, battery health, and overall DC system voltage.

Victron Configuration in Genesis

To enable **Victron Energy Monitoring** within *Genesis*, the system requires certain key inputs that allow it to securely connect to the Victron VRM portal and retrieve live operational data from the associated devices.

1. **Device Type:** `Victron`
2. **Device Name:** A valid and unique name for identification in Genesis.

3. **Serial Number:** The serial number of the device, obtainable from the **Victron VRM Portal**.

4. **Username and Password:** Credentials with administrative privileges to access or read the configured device in the VRM Portal.

Configuration Screenshot

Configuration Outcome

Once configured, **Genesis automatically detects** the available **PV Controller** and **Battery Manager**, initiating continuous polling of the **Victron API**. This process enables real-time retrieval of metrics such as battery voltage, charging current, solar power generation, and overall system status. These values are then continuously evaluated against the thresholds defined under the *Energy Monitoring* section of your alarm rules.

EFOY Fuel Cell Monitoring Configuration

EFOY devices provide **backup power generation** for sites where continuous power is critical. Their configuration ensures the fuel cell remains in optimal operating condition.

EFOY Configuration in Genesis

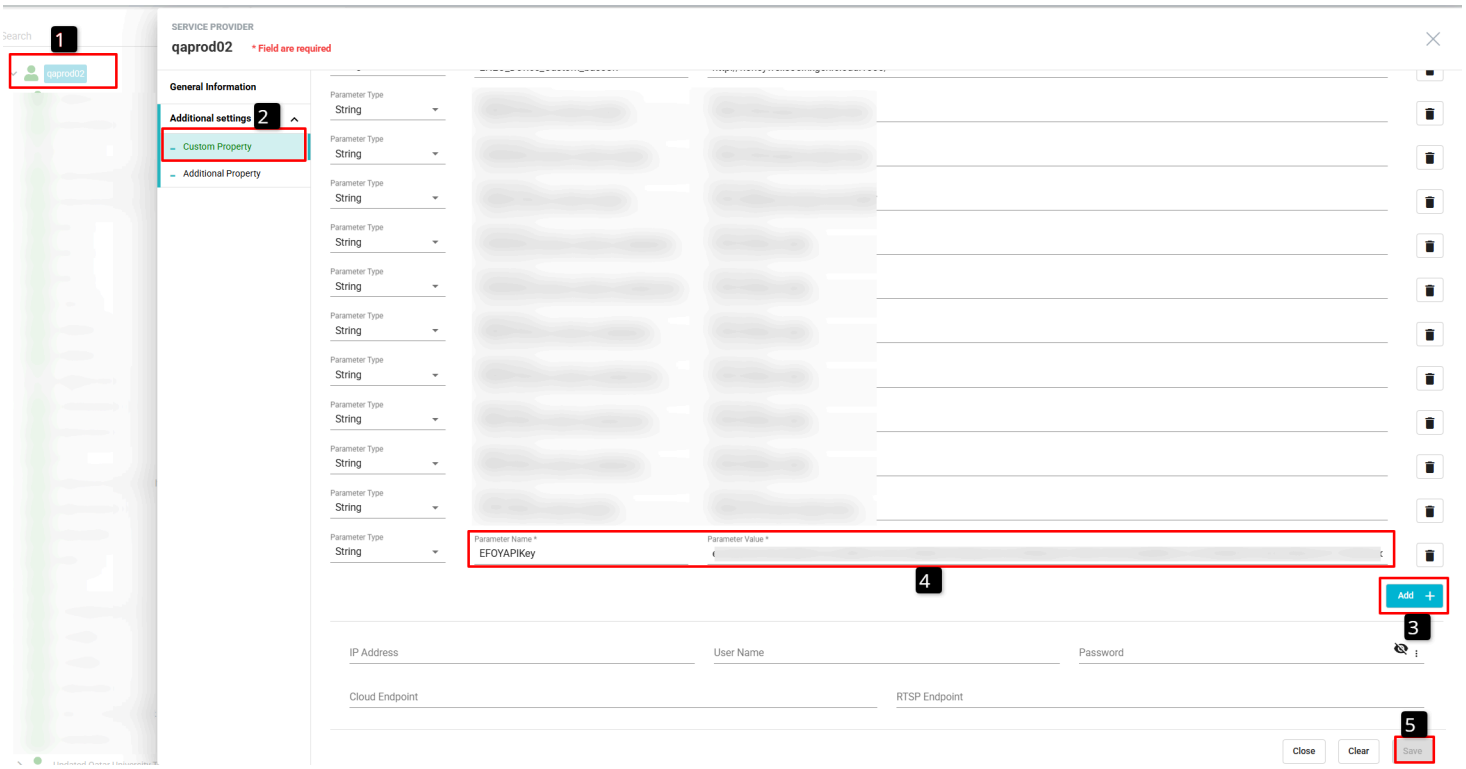
To enable **EFOY Fuell Cell** within *Genesis*, the system requires certain key inputs that allow it to securely connect to the EFOY Cloud portal and retrieve live operational data from the associated devices.

1. **Device Type:** `EFOY Cloud`

2. **Device Name:** A valid and unique name for identification in Genesis.

3. **Serial Number:** The unique serial number of the fuel cell, as listed in the **EFOY Cloud Portal**.

4. **EFOY API Key:** A valid API key with sufficient privileges to access all **Solar** and **Power Controller** parameters. It must be configured at Tenant level custom settings. Please find the screenshot below:



a. Select the tenant

b. Click Edit

c. Goto Custom Property

d. Click Add

e. Name the property as "EFOYAPIKey" and provide the original API key as value.

f. Click Save.

Configuration Screenshot

In the **IP Address field**, you may optionally specify the **EFOY Cloud API endpoint** instead of a local device address. The **Username and Password fields** are optional. however, you can provide **valid EFOY Cloud credentials** as a backup to the API key**.**

Configuration Outcome

Once configured, Genesis connects securely to the **EFOY Cloud API** and begins fetching live operational data. This includes parameters such as **fuel cell voltage**, **output power**, **cartridge level**, **energy generation rate**, and **system health**—providing complete visibility into fuel cell performance and runtime efficiency.

Teltonika IoT Connectivity Configuration

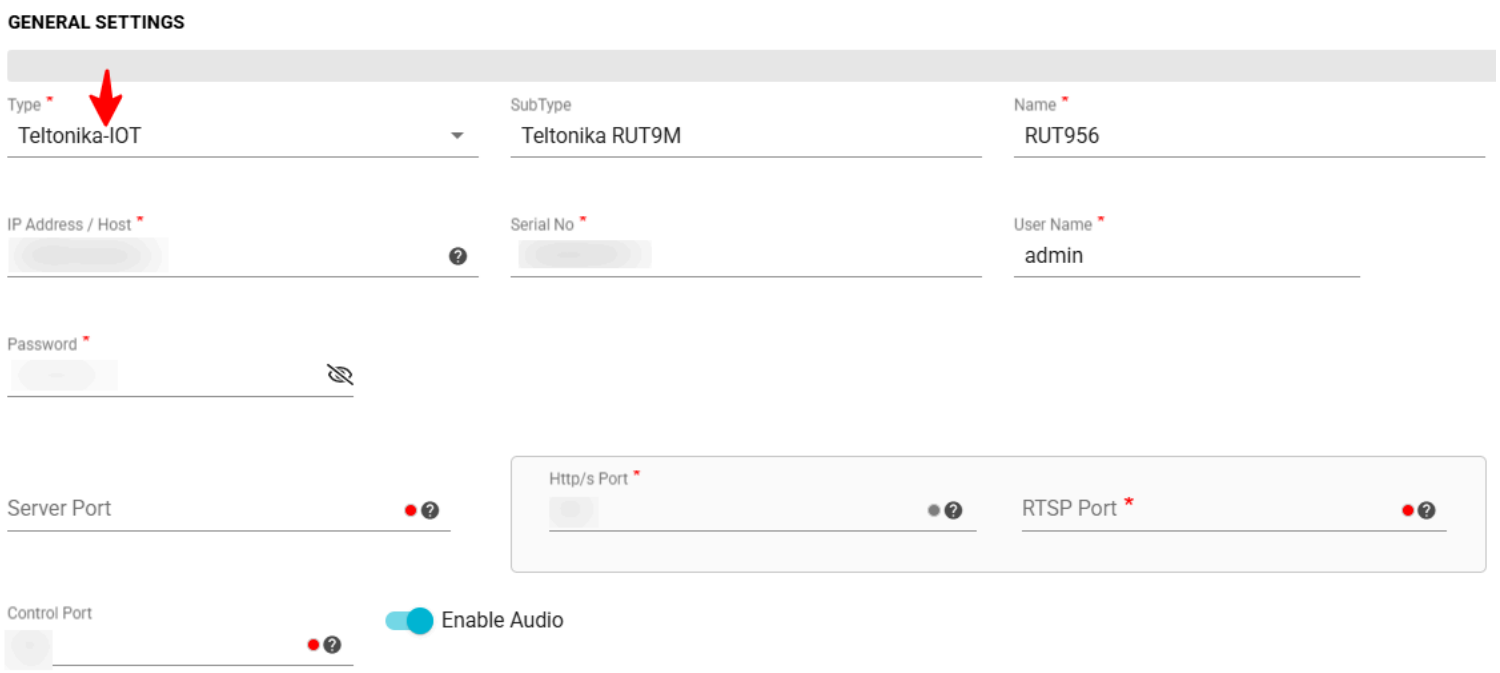
Teltonika devices manage **connectivity, communication, and environmental interaction** for each tower. They report detailed I/O status, network health, SIM usage, GPS, and signal jamming conditions.

Teltonika IOT Configuration in Genesis

To enable **Teltonika IOT Router** within *Genesis*, the system requires certain key inputs that allow it to securely connect to the router and retrieve live operational data from the associated devices.

1. **Device Type:** `Teltonika-IOT`
2. **Device Name:** A valid and unique name for identification in Genesis.
3. **IP Address:** The **public** or **VPN-accessible** IP address of the Teltonika device.
4. **Serial Number:** The unique serial number of the router.
5. **Control Port:** The valid HTTP port used for device communication.
6. **Username and Password:** Administrative credentials that allow full access to all device parameters and subcomponents.

Configuration Screenshot



Configuration Outcome

Once connected, Genesis establishes a secure communication channel with the **Teltonika Router**, enabling continuous monitoring of key metrics such as **signal strength**, **network uptime**, **connected I/O states**, and **environmental sensor data**. Any deviations—such as loss of connectivity, abnormal input voltage, or I/O threshold breach—are automatically logged and can trigger **real-time alerts** to CMS.

Alarm Rules Overview

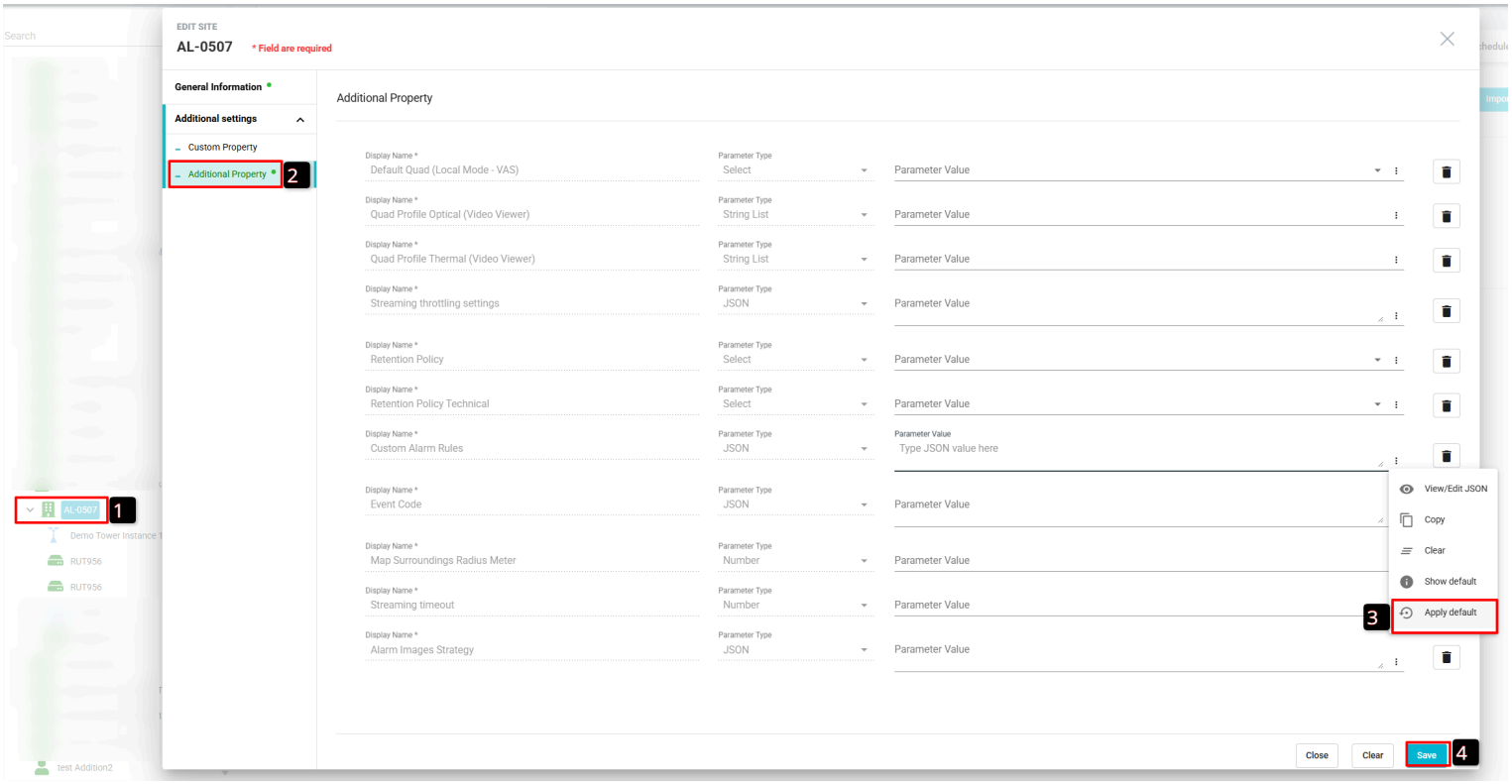
This section outlines the **structure, purpose, and usage** of the alarm rules configuration used for **remote tower monitoring** within Genesis.
Each part of the configuration defines a monitored category, its key parameters, and the conditions under which alerts are raised.

Configuring Alarm Rules

The **Alarm Rules** in Genesis are defined in **JSON format**, allowing precise and flexible configuration of alerting behavior.
Rules can be configured at either the **Site level** or **Device level**, depending on the monitoring scope.

To configure alarm rules:

1. Navigate to the desired **Site** or **Device** in Genesis.
2. Click **Edit** and go to the **Additional Properties** section.
3. Locate the property named **Custom Alarm Rules**.
4. Open the property's **hamburger menu** and select **Apply Default** to load the preconfigured default rule set.
    - By default, all rules are **inactive**.
5. Modify the JSON configuration to **enable specific rules** by setting the `"active"` parameter to `true` for the relevant parameters.
6. Once saved, Genesis automatically **applies the configured thresholds**, continuously evaluates incoming telemetry, and **triggers alerts** when any parameter exceeds or falls below its defined range.



The **Custom Alarm Rules** configuration JSON in Genesis is structured into three main sections: **Energy Monitoring**, **Fuel Cell Monitoring**, and **Connectivity Monitoring**.

- The **Energy Monitoring** section contains alert configurations specific to **Victron** devices, covering parameters such as battery voltage, and state of charge.
- The **Fuel Cell Monitoring** section defines alert rules for **EFOY** systems, including fuel level, voltage, temperature, and state of charge thresholds.
- The **Connectivity Monitoring** section manages alerts for **Teltonika IoT** devices, covering aspects such as network signal quality, SIM data usage, input/output status changes, and jammer detection.

This modular structure allows each device type to be independently configured while maintaining a unified monitoring and alerting framework across the entire tower system.

Victron Energy Monitoring

Battery Voltage Rule

Monitors the solar or battery voltage levels.

Default JSON Configuration

```
      "voltage": {
  "active": false,
  "low": 12,
  "high": 14,
  "eventCode": "battery.voltage.warning",
  "groupCode": "tower.solar.voltage.alert"
}
```

Configuration Parameters

Parameter Name	Description	Example Value
`active`	Enables or disables this specific alarm rule. Set to `true` to activate.	`true`
`low`	The lower voltage threshold. An alert triggers if the voltage falls below this value.	`12`
`high`	The upper voltage threshold. An alert triggers if the voltage exceeds this value.	`14`
`eventCode`	Internal event identifier used for integration with CMS or external notification systems.	`battery.voltage.warning`
`groupCode`	Group identifier used by Genesis Tower Alarm Manager to categorize alerts.	`tower.solar.voltage.alert`

Example Behavior

- When voltage falls **below 12V**, an alert with event code `battery.voltage.warning` is raised.
- When voltage returns **within range (12V–14V)**, Genesis automatically logs the recovery event and clears the alert.
- When voltage exceeds **14V**, a high-voltage warning is generated.

Battery State of Charge (SOC) Rule

Monitors the remaining charge percentage in the battery bank.

Default JSON Configuration

```
    "stateOfCharge": {
  "active": false,
  "low": 15,
  "eventCode": "battery.soc.warning",
  "groupCode": "tower.battery.soc.alert"
}
```

Configuration Parameters

Parameter Name	Description	Example Value
`low`	Minimum acceptable SOC percentage	`15`

Example Behavior

- SOC below **15%** triggers low battery alert.
- Restores to normal when SOC rises above the threshold.

EFOY Fuel Cell Monitoring

Fuel Level Rule

Monitors the available fuel percentage in the EFOY fuel cartridge.

Default JSON Configuration

```
    "fuelLevel": {
  "active": false,
  "low": 15,
  "eventCode": "fuel.low.warning",
  "groupCode": "tower.fuel.level.alert"
```

```
}
```

Configuration Parameters

Parameter Name	Description	Example Value
`low`	Minimum acceptable Fuel level	`15`

Example Behavior

- Triggers a warning when fuel drops below **15%**.

Voltage Rule

Monitors the EFOY power output voltage.

Default JSON Configuration

```
      "voltage": {
  "active": false,
  "low": 12,
  "high": 15,
  "eventCode": "fuel.voltage.critical",
  "groupCode": "tower.fuel.voltage.alert"
}
```

Example Behavior

- When voltage falls **below 12V**, an alert with event code `fuel.voltage.warning` is raised.
- When voltage returns **within range (12V–14V)**, Genesis automatically logs the recovery event and clears the alert.
- When voltage exceeds **14V**, a `fuel.voltage.warning` warning is generated.

State of Charge Rule

Tracks the internal battery's charge status in EFOY.

Default JSON Configuration

```
    "stateOfCharge": {
  "active": false,
  "low": 15,
  "eventCode": "fuel.soc.warning",
  "groupCode": "tower.fuel.soc.alert"
}
```

Example Behavior

- Triggers an alert when SOC drops below **15%**.

Temperature Rule

Monitors EFOY system temperature.

Default JSON Configuration

```
    "temperature": {
  "active": false,
  "low": 10,
  "high": 35,
  "eventCode": "fuel.temperature.warning",
  "groupCode": "tower.fuel.temperature.alert"
```

}

Example Behavior

- Below **10°C** or above **35°C** → Thermal warning.

Teltonika IoT Connectivity

Input/Output Status Change Rule

Monitors physical input/output state changes in Teltonika I/O ports.

Default JSON Configuration

```
      "IOStatusChange": {
  "input": {
    "AnalogCurrentLoop": {"active": false, "low": 4, "high": 20},
    "AnalogInput": {"active": false, "low": 11, "high": 14},
    "DigitalInput": {"active": false, "ranges": ["Low level", "High level"], "alertOn": "High Level"},
    "IsolatedInput": {"active": false, "ranges": ["Low level", "High level"], "alertOn": "High Level"},
    "PowerSocketInput": {"active": false, "ranges": ["Low level", "High level"], "alertOn": "High Level"},
    "eventCode": "input.statechange",
    "groupCode": "tower.input.statechange"
  },
  "output": {
    "IsolatedOutput": { "active": false, "ranges": ["Low level", "High level"], "alertOn": "High Level" },
    "Relay": { "active": false, "ranges": ["open", "closed"], "alertOn": "open" },
    "PowerSocketOutput": { "active": false, "ranges": ["Low level", "High level"], "alertOn": "High Level" },
    "eventCode": "output.statechange",
    "groupCode": "tower.output.statechange"
  }
}
```

Analog Current Loop

Field	Description
active	Enables 4–20 mA loop monitoring (true/false)
low / high	Defines acceptable current range.
eventCode	`input.statechange`
groupCode	`tower.input.statechange`

Analog Input

Field	Description
active	Enables analog voltage input monitoring (true/false)
low / high	Acceptable range (e.g., 11–14 V).
eventCode	`input.statechange`
groupCode	`tower.input.statechange`

Digital / Isolated / Power Socket Inputs

Field	Description
active	Enables input monitoring (true/false)
ranges	Possible states, e.g., `["Low level", "High level"]`.
alertOn	State that should trigger an alarm (e.g., “High Level”).
eventCode	`input.statechange`
groupCode	`tower.input.statechange`

Isolated / Power Socket Outputs

Field	Description
active	Enables Output monitoring (true/false)
ranges	Possible states, e.g., `["Low level", "High level"]`.
alertOn	State that should trigger an alarm (e.g., “High Level”).
eventCode	`output.statechange`
groupCode	`tower.output.statechange`

Relay Output

Field	Description
active	Enables relay trigger monitoring (true/false)
ranges	Possible states, e.g., `["open", "closed"]`.
alertOn	Tracks relay states (e.g., “open”).
eventCode	`output.statechange`
groupCode	`tower.output.statechange`

SIM Card Data Package Usage Rule

Tracks mobile data consumption for Teltonika routers.

Default JSON Configuration

```
    "simCardDataPackage": {
  "active": false,
  "simBillingStartDate": 1,
  "simDataPackageGB": 10,
  "usageWarningPercent": 80,
  "eventCode": "sim.data.usage.warning",
  "groupCode": "tower.connectivity.simdata.alert"
}
```

Configuration Parameters

Parameter Name	Description	Example Value
`simBillingStartDate`	Billing cycle start date of the SIM plan **(currently inactive)**	`1`
simDataPackageGB	Total mobile data allocated per month **(in GB)**	10
usageWarningPercent	Warning threshold as a percentage of total data usage	80

Example Behavior

- Triggers an alert when data usage exceeds **80% of 10GB**.

Cellular Network Loss Rule

Detects loss of mobile network connectivity.

Default JSON Configuration

```
    "cellNetworkLoss": {
  "active": false,
  "eventCode": "modem.signal.warning",
  "groupCode": "tower.connectivity.signal.alert"
}
```

Example Behavior

An alert is triggered when the network signal drops very low. Genesis automatically clears the alert when the signal back to acceptable range.

Geo-Fencing Rule

Monitors device location relative to its assigned site.

Default JSON Configuration

```
    "geoFencing": {
  "active": false,
  "radiusMetersWarning": 500,
  "eventCode": "geo.fence.breach",
  "groupCode": "tower.connectivity.geofence.alert"
}
```

Configuration Parameters

Parameter Name	Description	Example Value
`radiusMetersWarning`	**Radius (in meters) from the defined geo-fence center (typically, the site's geo location) within which the tower must remain**	`500`

Example Behavior

This rule continuously monitors the **GPS coordinates** reported by the Teltonika router or GPS module.
If the device moves **outside the configured radius**, Genesis generates an alert indicating a potential **tower displacement or unauthorized movement**.
The **location boundary** is defined by the **configured coordinates of the parent Site**, ensuring that alerts are triggered only when the tower moves beyond its designated operational area.

The alert is automatically cleared when the tower returns within the defined site boundary.

Jammer Detection Rule

Default JSON

```
    "jammerDetection": {
  "active": false,
  "criteria": {
    "signalDropThreshold": 25,
    "minSignalLevel": -113,
    "rsrqThreshold": -20
  },
  "eventCode": "modem.jammer.detected",
  "groupCode": "tower.connectivity.jamming.alert"
}
```

Configuration Parameters

Parameter Name	Description	Example Value
signalDropThreshold	Percentage drop in signal strength considered as a potential jamming event	25
minSignalLevel	Minimum acceptable signal level (in dBm); if signal goes below this, jamming is suspected	-113
rsrqThreshold	Minimum acceptable RSRQ (Reference Signal Received Quality); poor RSRQ may indicate interference	-20

Example Behavior
This rule detects **potential network jamming attempts** based on sudden signal degradation and poor RSRQ quality.
If the **signal drop percentage** exceeds the configured threshold or the **RSRQ value** falls below the acceptable range, Genesis triggers a **jamming alert**.
The alert clears automatically once normal signal stability is restored.

Configuration Notes

- Each rule has an **`active` flag**. When `false`, the system skips that rule.
- **Thresholds** can be adjusted per site depending on environmental conditions.

- **Event Codes** are unique identifiers used internally within Genesis for system integrations — particularly for notifying external **Central Monitoring Systems (CMS)** or third-party applications. They ensure consistent and standardized communication of alarm events across different platforms.
- **Group Codes**, while similar in structure to Event Codes, are specifically utilized within the **Tower Monitoring Environment**. They serve as logical identifiers for grouping related alarms, enabling efficient organization, filtering, and management of events within Genesis and its tower-level alerting workflows.